HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed Python-based microservices using **FastAPI** to serve model inference endpoints, handling an average of **~4,500 requests per hour**.
- Engineered a **Retrieval-Augmented Generation (RAG) agent** by integrating a vector database (Pinecone) with an LLM API (OpenAI GPT-4) to provide contextually relevant answers from a corpus of 10,000+ documents.
- Designed and implemented a **LangChain-based workflow** for prompt engineering and output validation, enabling iterative refinement of LLM prompts for a customer-facing Q&A; system.
- Contributed to the development of a **Java Spring Boot** application for internal tooling, implementing RESTful APIs for data retrieval and processing.
- Containerized microservices and data pipelines using **Docker**, facilitating consistent deployment across development and staging environments on **AWS EC2**.
- Implemented robust error handling and logging mechanisms within Python applications, integrating with **Splunk** for centralized monitoring and analysis of production systems.
- Collaborated with cross-functional teams to define technical requirements and conduct code reviews for new feature development, ensuring adherence to engineering best practices.

SKILLS

AI/ML – LLM APIs, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Vector Databases, FastAPI, scikit-learn, OpenAI GPT-4, Generative AI

Backend Development – Python, Java, Spring Boot, REST APIs, Microservices, Flask, Docker, AWS

Data Engineering & Databases – SQL, PostgreSQL, Oracle, MySQL, Data Pipelines, ETL

DevOps & CI/CD – Git, Jenkins, Docker, AWS EC2, Splunk

Analytics & Visualization – Power BI, Tableau, DAX, R, pandas, NumPy

PROJECTS

Customer Q&A; System with RAG

- Developed a **Python application** integrating **LangChain** and **OpenAI's GPT-4 API** to create a conversational Q&A; interface for enterprise customers.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** by embedding documents into a **Pinecone vector database**, improving response accuracy by **~30%**.
- Designed and executed test cases to evaluate LLM response quality, focusing on prompt effectiveness and identifying potential hallucinations.

AI-Powered Data Analysis Tool

- Built a **FastAPI microservice** in Python to process and analyze large datasets, providing insights through automated reporting.
- Integrated a **scikit-learn classification model** into the pipeline to predict customer churn based on historical transaction data, achieving **~0.82 AUC** on a held-out test set.
- Containerized the application using **Docker** for simplified deployment and scaling on cloud infrastructure.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)